Quantum Information - Formula Sheet

1 Lecture 1

Fundamental Thermodynamics & Equations of State

- **Ideal Gas Law:** $PV = Nk_BT = nRT$
- **First Law (Finite):** $\Delta U = \Delta Q - \Delta W$
- **First Law (Infinitesimal):** $dU = \delta Q - \delta W = TdS - PdV = dU = \sum_j \epsilon_j dn_j + \sum_j n_j d\epsilon_j$
- **Work:** $dW = PdV$
- **Heat Capacity:** $\Delta Q = C\Delta T$

Heat Engines & Second Law

- **Work in Cycle:** $W = Q_2 - Q_1$
- **Engine Efficiency:** $\eta = \frac{W}{Q_2} = 1 - \frac{Q_1}{Q_2}$
- **Clausius Theorem:** $\oint \frac{dQ}{T} \leq 0$; if equal then reversible

Macroscopic Entropy

- **Entropy Change (Reversible, from Clausius):** $\Delta S = \int_{B;rev}^{A} \frac{dQ}{T} \leq S(A) - S(B)$
- **Third Law:** $S = 0$ at $T = 0$; $S(A) = \int_0^{T_A} C_R \frac{dT}{T}$; $S(A) \to 0$ as $T_A \to 0$

Boltzmann Distribution

- **Internal Energy:** $U = \sum_j n_j \epsilon_j$
- **Total Particles:** $N = \sum_j n_j = A \sum_j \exp(\beta \epsilon_j)$
- **Multiplicity:** $t(\{n_j\}) = \frac{N!}{\prod_j n_j!} \prod_j g_j^{n_j} \propto \Omega\{n_j\}$
- **Stirling's Approximation:** $\ln n! \approx n \ln n - n$

- **Boltzmann Distribution:** $n_j^* = \frac{N}{Z} \exp(\beta \epsilon_j) = \exp(\alpha + \beta \epsilon_j)$
- **Partition Function:** $Z = \sum_j \exp(\beta \epsilon_j)$
- **Occupation Numbers:** $n_j = \frac{N}{Z} \exp(\beta \epsilon_j)$
- **Lagrange Multiplier:** $\beta = -\frac{1}{k_B T}$

Microscopic Entropy

- **Boltzmann Entropy:** $S = k_B \ln \Omega = -\beta T \ln(\Omega)$
- **Gibbs Entropy:** $S = -k_B \sum_{j=1}^{m} P_j \ln P_j$; $P_j = \frac{n_j}{N}$

Energy Decomposition

- **Heat Component:** $\sum_j \epsilon_j dn_j$ (redistribution of particles)
- **Work Component:** $\sum_j n_j d\epsilon_j$ (shifting energy levels)

Diffusion

- **Diffusion Equation:** $\frac{\partial n}{\partial t} = D \frac{\partial^2 n}{\partial x^2}$
- **Gibbs Entropy (Continuous):** $S = -c \int n(x,t) \ln n(x,t) dx$
- **Entropy Derivative:** $\frac{dS}{dt} = cD \int \frac{1}{n} \left(\frac{\partial n}{\partial x}\right)^2 dx > 0$
- **Gaussian Solution:** $P(x,t) = \frac{1}{\sqrt{4\pi D t}} \exp\left(-\frac{x^2}{4Dt}\right)$
- **Resulting Entropy:** $S(t) \sim \log(4\pi D t)$ (grows logarithmically with time)

2 Lecture 2

Fundamental Measures of Information

- **Information Measure:** $f(N) = K \log N$ for N equally likely messages
- **Information Units:** Base 2 (bits), Base 10 (dits/Hartleys), Base e (nats), Base 256 (bytes)
- **Change of Base:** Multiply by $\log_b a$ to convert from base a to base b

Discrete Noiseless Channel Capacity

- **Teletype Capacity:** $H = \log_2 N$ (bits per symbol); $C = n \log_2 N$ (bits per second)
- **General Channel Capacity:** $C = \lim_{T\to\infty} \frac{\log N(T)}{T}$
- **Varying Symbol Durations:** $X^{-t_1} + X^{-t_2} + \cdots + X^{-t_n} = 1$; $C = \log X_0$
- **State Constraints:** $\left|\sum_s W^{-b_{ij}^{(s)}} - \delta_{ij}\right| = 0$; $C = \log W$

Stochastic Sources and Markov Processes

- **Joint Probability:** $p(i,j) = p(i)p_i(j)$
- **Marginal Probability:** $p(i) = \sum_j p(i,j) = \sum_j p(j)p_j(i)$
- **Equilibrium Condition:** $P_j = \sum_i P_i p_i(j)$

Shannon Entropy

- **General Entropy:** $H = -K \sum_{i=1}^{n} p_i \log p_i$
- **Binary Case:** $H = -(p \log p + q \log q)$
- **Recursive Rule:** $H(p_1, \ldots, p_N) = H(z_1, \ldots, z_M) + \sum z_i H(d_{i,1}, \ldots, d_{i,r_i})$
- **Entropy per Second:** $H' = mH$ (where m is average symbols per second)

Joint and Conditional Entropy

- **Joint Entropy:** $H(x,y) = -\sum_{i,j} p(i,j) \log p(i,j)$
- **Conditional Entropy:** $H_x(y) = -\sum_{i,j} p(i,j) \log p_i(j)$
- **Chain Rule:** $H(x,y) = H(x) + H_x(y)$
- **Independence:** $H(x,y) \leq H(x) + H(y)$ (equality if independent)
- **Information Inequality:** $H(y) \geq H_x(y)$ (knowledge reduces uncertainty)
- **Limits:** $0 \leq H \leq \log n$ (maximized when all p_i equal)

Coding and Transmission

- **Fundamental Theorem:** Maximum transmission rate is C/H symbols per second
- **Coding Length:** $\log_2 \frac{1}{p_s} \leq m_s < 1 + \log_2 \frac{1}{p_s}$
- **Coding Efficiency:** $G_N \leq H_1 < G_N + \frac{1}{N}$
- **Redundancy:** $1 -$ (Relative Entropy)